Pascal's triangle

1 The 4th row of Pascal's triangle consists of the numbers $1, 4, 6, 4, 1$.

- **a** Write down the numbers in the 5th row of Pascal's triangle.
- **b** Find the sum of the entries in the 5th row of Pascal's triangle.
- **c** Rewrite each element of the 5th row of Pascal's triangle as an ${}^{n}C_{r}$ coefficient.

2 Complete the following entries in a particular row of Pascal's triangle:

$$1, 8, \square, 56, 70, \square, 28, \square, 1$$

3 Which row of Pascal's triangle would give us the coefficients of terms in the expansion of $(x+y)^7$?

4 State the number of terms in the following expansions:

- **a** $(4x+y)^6$
- **b** $(m+y)^8$

5 Consider the expansion of $(y+k)^4$.

- **a** How many terms will there be in the expansion?
- **b** Write the coefficient of the 2nd term.

6 Using the relevant row of Pascal's triangle, determine the coefficient of each term in the following expansion:

$$(x+m)^4 = \square x^4 m^0 + \square x^3 m^1 + \square x^2 m^2 + \square x^1 m^3 + \square x^0 m^4$$

Binomial theorem

7 Evaluate the following:

- **a** $\binom{6}{5}$
- **b** $\binom{14}{1}$
- **c** $\binom{10}{9}$
- **d** $\binom{8}{0}$

8 Find the specified term for the following expansions:

- **a** Fifth term of $(b+2)^7$.
- **b** Fifth term of $(c+d)^9$.
- **c** Fifth term of $(3u-v)^6$.
- **d** Tenth term of $\left(u+\dfrac{1}{2}\right)^{12}$.
- **e** Seventh term of $\left(\dfrac{3y}{2}-\dfrac{2}{3y}\right)^{10}$.
- **f** Eleventh term of $\left(2x+y^2\right)^{13}$.
- **g** Sixteenth term of $\left(x-y^5\right)^{19}$.
- **h** Middle term of $\left(4x^2+3y^3\right)^8$.
- **i** Middle term of $\left(-3x^{-3}+2y^{-2}\right)^4$.
- **j** Constant term of $\left(2y-\dfrac{1}{y^3}\right)^8$.

9 The first three terms of $(x+y)^{10}$ are x^{10}, $10x^9y$ and $45x^8y^2$. Using the symmetry of the coefficients, write the last three terms of the expansion.

10 In the expansion of $(m+k)^n$, the coefficient of the 9th term is 45.

- **a** Find the value of n.
- **b** Find the 9th term.

11 A particular term in the expansion of $\left(3a^2+\dfrac{p}{b}\right)^4$ is $\dfrac{96a^2}{b^3}$, for some constant p. Find the value of p.

12 Consider the expansion of $(a-4)^5$. Will the 2nd term be positive or negative?

13 By considering $(x+y)^4=(x+y)(x+y)(x+y)(x+y)$, in how many ways can three factors of x and one factor of y be chosen from the expansion to form the term which contains x^3y^1? Write your answer in the form ${}^nC_r$.

14 Complete the following expansion:

$$(4p+3q)^3={}^3C_0\,(4p)^{\square}\,(3q)^0+{}^3C_1\,(4p)^{\square}\,(3q)^1+{}^3C_2\,(4p)^{\square}\,(3q)^2+{}^3C_3\,(4p)^{\square}\,(3q)^3$$

15 Consider the binomial $(4x+3y)^4$.

- **a** State the first term in the expansion.
- **b** State the last term in the expansion.

16 Consider the expansion of $(a+b)^5$.

- **a** State the coefficient of the 3rd term in the form $\binom{n}{r}$.
- **b** Evaluate the coefficient of the 3rd term.
- **c** By considering the symmetry of $\binom{n}{r}$, which other term has the same coefficient as the 3rd term?

17 Which two terms in the expansion of $(u +$ [illegible] ave a coefficient of $\binom{11}{9}$?

18 Find the coefficient of x^{17} in the expansion of $\left(3x^2 + 2x\right)^{11}$.

19 Use the binomial theorem to expand the following expressions:

- **a** $(1+x)^4$
- **b** $(2+b)^5$
- **c** $(p+q)^5$
- **d** $(c-d)^8$
- **e** $(u+4v)^6$
- **f** $(y+3)^4$
- **g** $(ka-b)^6$
- **h** $(y-5)^3$
- **i** $\left(y-\frac{1}{3}\right)^3$
- **j** $\left(2+\frac{y}{2}\right)^3$
- **k** $(3y+2r)^3$
- **l** $\left(a-\frac{1}{a}\right)^3$
- **m** $\left(u^2+3v^2\right)^3$
- **n** $\left(1+\frac{4}{y}\right)^3$

20 Find:

- **a** The term in $(3+\sqrt{x})^{10}$ that contains x^4.
- **b** The term in $(c+d)^{10}$ that contains c^3.
- **c** The term in $(u+v)^{11}$ that contains v^8.
- **d** The term in $(u+v)^{15}$ that contains u^5.

21 If the eighth and tenth terms in the expansion of $(x+y)^n$ have the same coefficients, find the value of n.

22 Find the coefficient of a^3b^6 in the expansion of $\left(2a-\frac{b}{2}\right)^9$.

23 Consider the expansion of $(1+x)^9$. State the coefficient of the 4th term in the form $\binom{n}{r}$.

24 Find the value of n such that the expansion of $(a+b)^n$ contains the term $84a^6b^3$.

25 Complete the following expansion by determining the missing power and binomial coefficient of the $(k+1)$th term:

$$(3-x)^8 = {}^8C_0 \times 3^8(-x)^0 + {}^8C_1 \times 3^7(-x)^1 + \ldots + \square \times 3^{8-k}(-x)^{\square} + \ldots + {}^8C_8 \times 3^0(-x)^8$$

26 Consider the binomial series:

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots$$

- **a** Use the first four terms to approximate $(1.03)^{-2}$ to the nearest thousandth.
- **b** Use the first four terms to approximate $(1.05)^{\frac{3}{4}}$ to the nearest thousandth.
- **c** Use the first five terms to approximate $\frac{1}{(1.06)^2}$ to the nearest thousandth.